

Freight Rate Prediction

Data validation, modelling approach and final predictions

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Task	Predict posted_rate for 12,000 unseen loads
Final model	CatBoostRegressor — 18 features, chronological validation

October 1, 2026

1. Executive Summary

The objective is to predict posted freight rates from shipment, geographic, equipment, market and temporal information. The workflow covers data-quality validation, chronological holdout design, categorical modelling with CatBoost, feature engineering experiments, model comparison, error analysis and generation of predictions for 12,000 unlabeled loads.

The selected model is a CatBoostRegressor trained directly on posted_rate using 18 features. On the 9,523-row internal validation period, it achieved MAE of \$111.62, RMSE of \$632.73 and R² of 0.8281, with early stopping selecting iteration 154.

MAE	RMSE	R ²
\$111.62	\$632.73	0.8281

Key findings

- Distance is the dominant predictive variable, with 86.54 importance points in the CatBoost importance output.
- The chronological holdout contains 38,477 training observations from January–August 2025 and 9,523 validation observations from September–October 2025.
- A log-transformed target and five additional engineered numerical features were tested, but both variants scored slightly worse than the 18-feature baseline.
- The main remaining weakness is the small group of extreme-rate loads: 24 validation records had actual rates of at least \$10,000 and were substantially under-predicted.
- The final prediction set contains 12,000 November–December 2025 loads; the notebook reports zero missing model inputs after cleaning.

2. Data Overview

Dataset	Rows	Columns	Role / period
train_test.csv	48,000	14	Labeled development • Jan 1–Oct 31, 2025
Internal training split	38,477	18 model inputs	Jan 1–Aug 31, 2025
Internal validation	9,523	18 model inputs	Sep 1–Oct 31, 2025
validation.csv	12,000	13 raw columns	Final prediction set • Nov 1–Dec 31, 2025
december_chart_inputs.csv	31	7 raw columns	Fixed December scenario

The development data contains shipment identifiers, pickup and delivery locations, geographic coordinates, distance, equipment, weight, date, market_index, quote_signal and the posted_rate target. The notebook reports 64 distinct locations and no duplicate rows.

The target is right-skewed. In the 48,000-row development data, posted_rate has mean \$2,373.98, median \$2,030.76, standard deviation \$1,486.49, minimum \$57.22 and maximum \$25,533.00.

3. Data-Quality Assessment

Issue	Development	Final validation	Treatment
Missing weight	300	310	Training-set median imputation
Missing market_index	374	249	Training-set median imputation
Negative weight	292	145	Set to missing, then impute
Duplicate rows	0	0	No duplicate rows

Invalid negative weights were treated as missing because physical shipment weight cannot be negative. The cleaning logic was applied consistently to development and unseen prediction data, and the final model inputs reported zero remaining missing values.

4. Validation Strategy

A chronological split was used instead of a random split. The model was trained on January–August 2025 and evaluated on September–October 2025. This preserves the time direction of the forecasting task and prevents later observations from being used to train the model before they are evaluated.

Training	Holdout	Final prediction
Jan–Aug 2025 38,477 rows	Sep–Oct 2025 9,523 rows	Nov–Dec 2025 12,000 rows

5. Feature Engineering

The final selected model uses 18 inputs. Categorical variables are handled natively by CatBoost.

Group	Features	Purpose
Categorical	pickup, delivery, equipment, route	Lane and equipment effects
Geographic	pickup/delivery latitude & longitude, distance	Route geography and length
Shipment	weight	Load size
Market	market_index, quote_signal	Market and quote conditions
Temporal	year, month, day, day_of_week, day_of_year, week_of_year	Calendar effects

A separate 23-feature experiment added weight_per_mile, distance_squared, weight_distance_interaction, latitude_difference and longitude_difference. These engineered features were evaluated separately and did not improve holdout performance.

6. Model Development and Selection

CatBoost was selected because the problem contains several categorical variables and nonlinear relationships. The final baseline was trained directly on posted_rate with RMSE as the evaluation metric and early stopping on the chronological holdout.

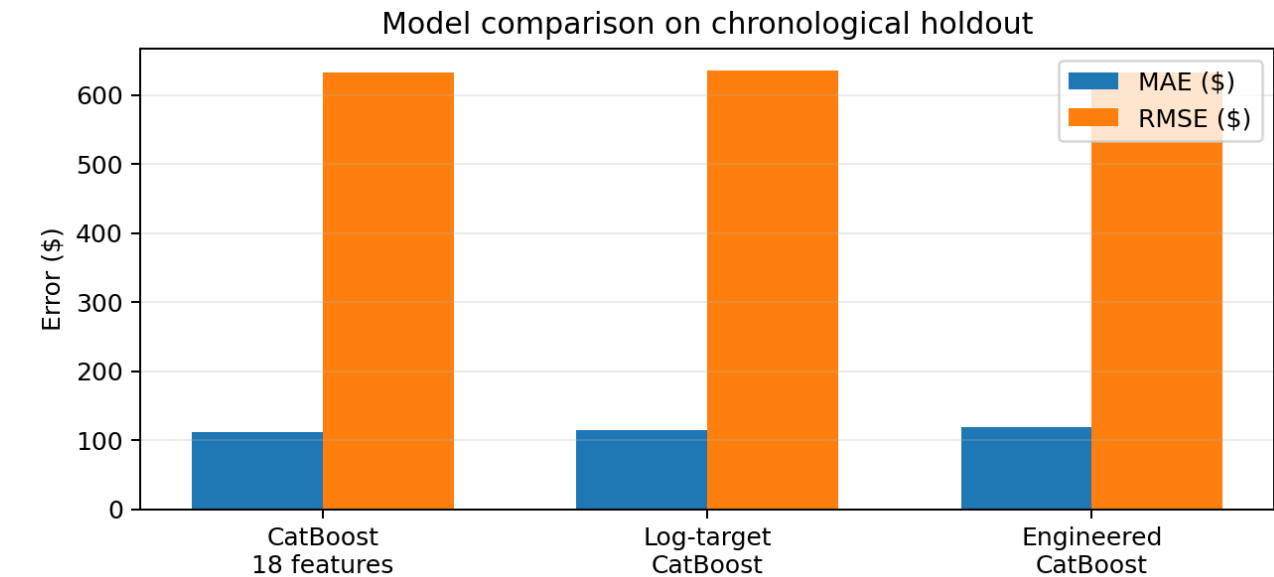


Figure 1. Comparison of the three CatBoost configurations on the September–October holdout.

Model	MAE	RMSE	R ²	Best iteration
CatBoost — 18 original features	\$111.62	\$632.73	0.8281	154
CatBoost — log target	\$114.52	\$635.84	0.8264	203
CatBoost — 23 engineered features	\$119.18	\$633.45	0.8277	180

The 18-feature direct-target model was retained because it produced the lowest MAE and RMSE and the highest R² among the tested candidates, while also being the simplest configuration.

7. Validation Results and Error Analysis

On the 9,523-row internal holdout, the final candidate achieved MAE \$111.62, RMSE \$632.73 and R² 0.8281. The difference between MAE and RMSE indicates that a relatively small number of large misses have a disproportionate effect on squared error.

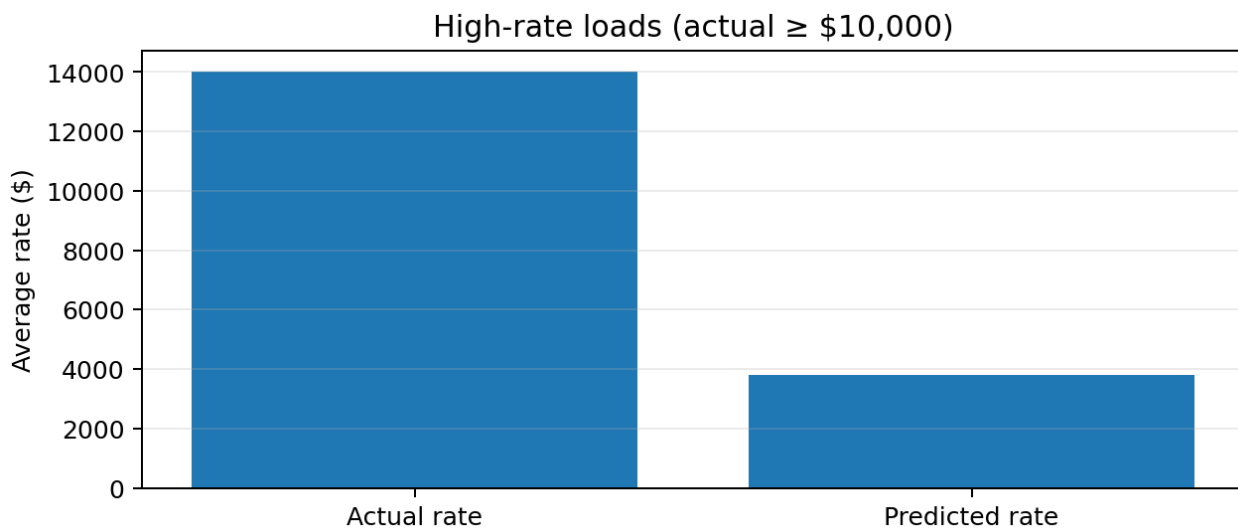


Figure 2. Average actual and predicted rates for validation loads with actual rate ≥ \$10,000.

There were 24 high-rate validation loads with actual rates of at least \$10,000. Their average actual rate was \$14,003.43, while the average predicted rate was approximately \$3,793.49, producing an average absolute error of approximately \$10,209.95. These records were evenly split between September and October; 16 used Dry Van equipment, 6 Reefer and 2 Flatbed.

Most of these extreme loads had distances between 1,000 and 3,000 miles. Their feature values generally resembled ordinary loads, so the available inputs provided limited evidence for the unusually high posted rates.

8. Feature Importance

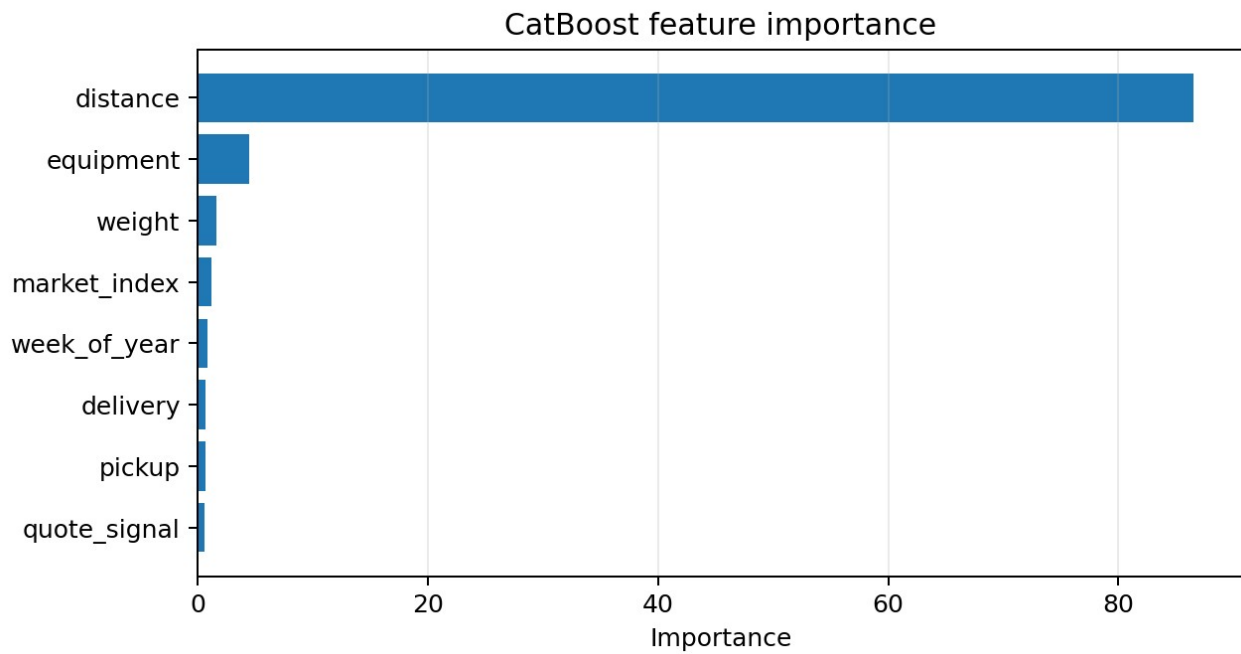


Figure 3. Top CatBoost feature-importance values for the final 18-feature model.

Rank	Feature	Importance
1	distance	86.54
2	equipment	4.46
3	weight	1.61
4	market_index	1.24
5	week_of_year	0.82
6	delivery	0.69
7	pickup	0.67
8	quote_signal	0.57

Distance is by far the strongest feature in the model, followed by equipment, weight and market_index. The result indicates that route length carries most of the predictable signal in the available data.

9. Stability Across Validation Months

Month	Observations	MAE	RMSE	Mean error
September	4,670	\$117.68	\$616.23	\$47.45
October	4,853	\$120.63	\$649.58	\$49.11

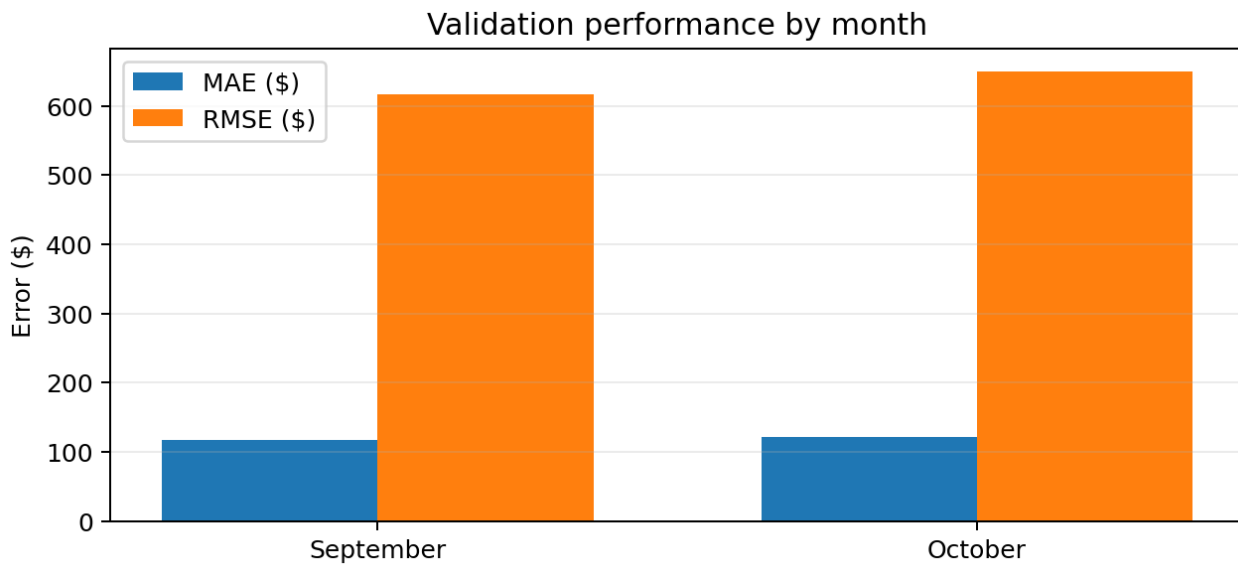


Figure 4. Monthly MAE and RMSE on the internal chronological holdout.

The model's monthly performance is relatively consistent across the two holdout months. October has somewhat higher RMSE, reflecting larger individual misses, while MAE remains close to September.

10. Final Prediction Generation

The final validation dataset contains 12,000 unlabeled loads. After the same cleaning, date-feature creation and route construction, the input is aligned to the selected 18-feature CatBoost model.

Check	Result	Status
Prediction rows	12,000	Passed
Output columns	load_id, predicted_rate	Passed
Missing predictions	0	Passed
Duplicate load IDs	0	Passed
Negative predictions	0	Passed

The final checked notebook reports 12,000 predictions with mean \$2,358.36, median \$2,041.51, minimum \$300.75 and maximum \$6,613.06. The output file contains exactly load_id and predicted_rate.

11. December Prediction Scenario

The supplied December scenario fixes the route as Lexington → Fort Wayne, distance at 360 miles, Dry Van equipment and 32,000 lb weight. The prediction changes across the 31 December dates.

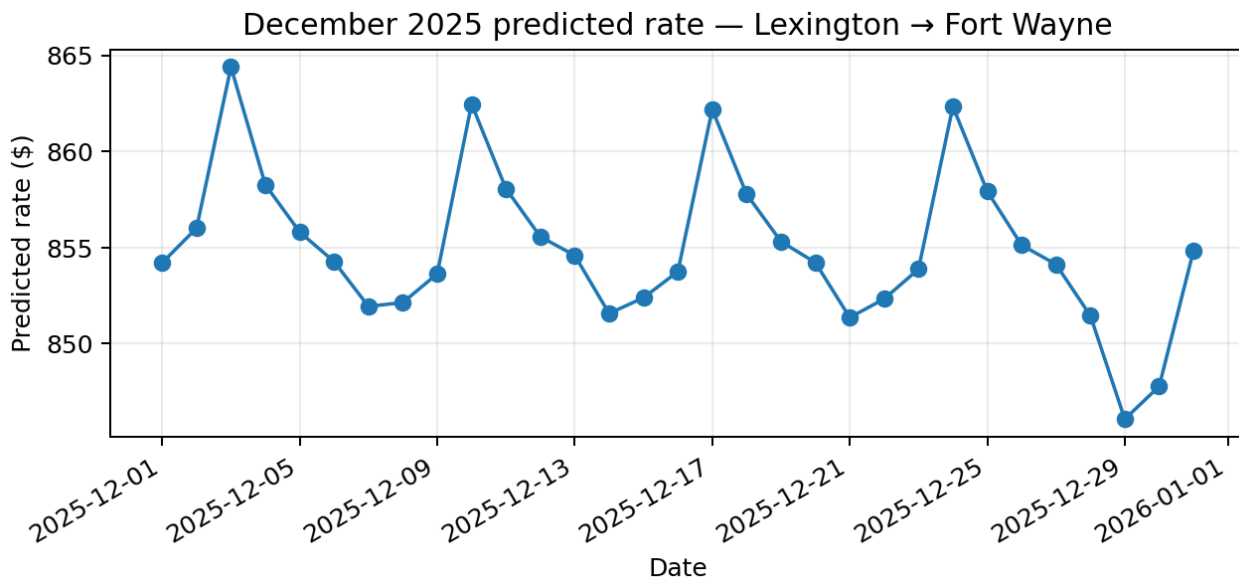


Figure 5. December 2025 predicted rate for the fixed Lexington → Fort Wayne scenario.

The chart shows a relatively narrow predicted range with recurring day-to-day variation across December. This scenario provides a reproducibility check because the route, distance, equipment and weight are fixed while the date changes.

12. Reproducibility and Submission Checklist

- Completed notebook: `freight_rate_ml_assessment_final_checked.ipynb`.
- Final prediction output: `validation_predictions.csv` with `load_id` and `predicted_rate`.
- Chronological validation: January–August training and September–October holdout.
- Final model: `CatBoostRegressor` with 18 features and early stopping at iteration 154.
- Data cleaning: invalid negative weights converted to missing and numerical gaps filled using training-set medians.
- December scenario visualization included in the report.
- Repository should include the notebook, prediction CSV, requirements/run instructions and assessment-provided scoring artifacts.

13. Limitations and Next Steps

- Refit on all labeled data after model selection so the final forecasting model can use the latest labeled observations.
- Use rolling-origin or expanding-window validation to estimate performance variability more robustly.
- Investigate the rare \$10,000+ loads because they dominate RMSE and appear poorly explained by the available features.
- Consider prediction intervals or quantile models to communicate uncertainty.
- Monitor market and route drift as November–December actual outcomes become available.

14. Conclusion

The completed workflow provides a reproducible freight-rate regression pipeline with explicit data-quality checks, chronological validation, native categorical modelling, model comparison, error analysis and final

prediction generation. The selected 18-feature CatBoost model achieved MAE \$111.62, RMSE \$632.73 and R^2 0.8281 on the September–October holdout.

The principal modelling limitation is the inability of the available features to explain the small set of extreme high-rate loads. The final prediction artifact contains 12,000 load-level rate estimates for the November–December 2025 prediction set with the required two-column output structure.